

DE QUEEN SEVIER COUNTY AP, AR, USA

WMO: 743312

Lat: 34.047N

Lon: 94.399W

Elev: 355

StdP: 14.51

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53925

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.5	23.7	8.9	8.8	29.8	13.5	11.0	33.4	18.7	53.5	16.7	50.4	1.9	340	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.3	98.8	76.2	96.0	76.5	93.4	76.5	80.4	90.0	79.5	89.6	78.5	88.5	6.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.1	148.1	84.0	77.0	142.3	83.6	75.9	137.2	82.8	44.2	89.7	43.1	89.3	42.2	88.5	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	12.9	11.4	13.5	102.2	3.9	4.2	10.6	105.2	8.3	107.7	6.2	110.1	3.3	113.1		
			12.6	82.1	3.7	1.5	10.0	83.2	7.8	84.1	5.7	84.9	3.0	86.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	42.6	45.8	54.8	62.0	70.3	78.6	81.3	81.8	75.3	63.7	53.0	44.9
	DBStd	15.76	9.21	9.26	8.79	7.22	6.32	3.73	3.74	4.11	5.66	8.30	8.90	8.73
	HDD50	785	266	176	52	6	0	0	0	0	0	5	69	211
	HDD65	2856	695	538	330	138	27	0	0	0	3	127	373	625
	CDD50	5503	37	59	202	365	630	857	971	986	759	428	157	52
	CDD65	2102	2	2	15	47	191	407	506	521	312	86	11	2
	CDH74	22137	4	14	145	506	1742	4224	5593	5781	3156	875	90	7
	CDH80	10396	0	1	11	76	536	1950	2930	3117	1492	271	12	0
Wind	WSAvg	4.0	4.4	5.0	5.3	5.0	4.1	3.6	3.4	3.1	3.2	3.5	3.9	4.2
Precipitation	PrecAvg	55.0	4.6	4.1	4.7	5.2	6.0	4.4	3.8	2.8	4.4	5.0	4.6	5.1
	PrecMax	74.1	7.9	9.0	10.6	11.6	13.6	9.2	8.6	8.1	10.3	12.9	13.7	9.0
	PrecMin	32.2	0.6	1.3	0.7	2.0	2.6	1.6	0.2	0.4	0.0	0.0	1.5	0.4
	PrecStd	10.9	2.0	1.9	2.5	2.4	2.8	2.2	2.4	1.7	2.4	3.6	3.1	2.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.5	76.3	81.6	85.7	91.0	97.0	101.6	104.4	98.7	90.1	81.4	74.6
		MCWB	60.4	63.1	65.3	68.9	73.6	75.8	76.2	75.7	74.1	72.3	68.6	65.3
	2%	DB	69.5	72.0	78.1	82.0	88.1	93.8	98.1	99.4	94.7	86.6	76.2	70.1
		MCWB	60.3	60.9	63.9	66.8	73.3	76.5	76.4	76.5	73.8	71.4	64.3	62.4
	5%	DB	65.6	68.1	74.8	79.6	85.6	91.5	95.1	96.8	91.5	82.9	72.4	66.3
		MCWB	59.2	59.0	62.7	66.8	72.8	76.5	77.3	76.9	74.2	68.7	63.9	59.2
	10%	DB	60.3	63.7	71.4	76.9	83.0	89.6	92.6	93.7	88.8	79.1	68.9	61.6
		MCWB	51.8	56.4	60.9	65.2	71.5	76.1	77.3	76.9	73.1	67.4	61.6	55.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.7	68.2	69.0	73.2	77.7	80.9	81.5	82.1	79.0	76.3	71.4	68.3
		MCDB	69.3	72.1	76.5	80.3	86.1	89.5	90.4	90.8	88.6	84.9	77.1	72.0
	2%	WB	63.7	64.6	67.1	70.9	76.0	79.4	80.3	80.4	77.7	74.6	68.6	64.7
		MCDB	67.7	68.7	74.6	78.3	84.3	88.8	90.3	90.1	88.1	82.4	73.3	68.9
	5%	WB	59.5	61.6	65.1	69.3	74.7	78.3	79.4	79.4	76.6	72.8	65.9	60.5
		MCDB	63.9	66.4	72.5	76.2	82.5	87.7	90.0	89.5	86.4	79.3	70.8	64.7
	10%	WB	53.8	57.2	62.8	67.6	73.2	77.3	78.4	78.4	75.2	70.2	63.0	56.4
		MCDB	57.9	61.5	69.4	74.1	80.6	86.5	88.5	88.7	84.1	75.6	67.6	60.0
Mean Daily Temperature Range	5% DB	MDBR	22.9	21.9	23.9	23.8	21.1	20.5	21.3	22.3	23.9	25.0	24.3	22.4
		MCDBR	29.4	29.5	26.8	26.5	23.5	23.4	25.5	27.3	27.1	28.2	27.4	28.1
		MCWBR	19.6	19.6	15.0	13.8	10.9	9.1	7.6	7.7	9.4	13.5	17.0	19.4
	5% WB	MCDBR	22.2	23.3	22.0	20.5	19.6	20.1	21.0	21.9	22.2	21.5	21.8	22.1
		MCWBR	17.9	17.7	13.7	12.8	10.2	8.7	8.2	8.3	8.8	11.6	16.3	17.8
Clear-Sky Solar Irradiance	taub	0.324	0.330	0.361	0.396	0.430	0.451	0.469	0.468	0.405	0.360	0.337	0.319	
	taud	2.488	2.476	2.413	2.315	2.283	2.263	2.242	2.232	2.372	2.479	2.494	2.514	
	Ebn at Noon	281	292	290	283	273	266	260	259	272	278	274	276	
	Edh at Noon	27	30	35	41	43	43	44	44	36	30	27	25	
All-Sky Solar Radiation	RadAvg	826	943	1315	1628	1785	1996	1981	1846	1564	1221	876	715	
	RadStd	75	143	101	103	165	201	130	142	137	171	108	92	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	+1.93	+2.75	N/A	N/A	N/A	N/A
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page